



EASTERN MEDITERRANEAN UNIVERSITY
FACULTY OF ENGINEERING
DEPARTMENT OF MECHANICAL ENGINEERING

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Course Name: THERMODYNAMICS 1
Semester/ Year: Fall 2023-2024
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Assessment: Lab 2
Due Date:

LAB NAME: THE HEAT ENGINE

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SECTIONS:

No.	STUDENT OUTCOMES	COURSE LEARNING OUTCOMES	WEIGHT OF SECTION /100	MARKS OBTAINED	REMARKS
Report Presentation	1, 6	2, 3, 4, 5, 6	10		
Introduction and Procedure	1, 6	2, 3, 4, 5, 6	20		
Results and Calculations	1, 6	2, 3, 4, 5, 6	20		
Discussion & Conclusion	1, 6	2, 3, 4, 5, 6	30		
Answer to Question	1, 6	2, 3, 4, 5, 6	20		
					Marks for report format & referencing are inclusive in the marks of each section.
TOTAL: (Out of 100)					

Course Learning Outcomes		Student Outcomes						
		1	2	3	4	5	6	7
1								
2		X					X	
3		X					X	
4		X					X	
5		X					X	
6		X					X	
Weight of Student Outcomes		H					H	

Instructions:

- A. Report presentation:** Draw neat, labeled diagrams where necessary. Write relevant equations and your assumptions where necessary. Be clear and specific and include units. Give explanatory notes where necessary. Carry out each step explicitly. Provide references to the data/model/equations used. References should be from the textbook or from an authentic source.
- B. Submission:** Upload your report as a pdf file to the "Lab Work" section on TEAMS within the designated time. The file uploaded should be named as [MENG245-Lab1-Student ID-Student Name]. No late submissions will be accommodated.

ABSTRACT

We used the PASCO TD-8572 Heat Engine tool to study how heat engines work by changing temperatures. In our hands-on experiment, we figured out two types of work: mechanical and thermodynamic. The tool, which has a piston, air chamber, and hoses, helped us learn about heat engines in a practical way. We adjusted temperatures and watched how the piston moved, connecting what we learned in class with real actions. We drew our results on a Pressure-Volume diagram, showing a full cycle with different processes. We found that thermodynamic work was 7.4556008×10^{-3} joules, and mechanical work, considering special processes, was 7.4556×10^{-3} joules. Both positive values mean the system did work. This experiment helps us understand thermodynamics better and shows the tool is useful for more studies.

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1. INTRODUCTION

The PASCO TD-8572 Heat Engine/Gas Law Apparatus lets us explore how things get hot and do work. In this experiment we will play around with temperature to see how well the heat engine performs.

Heat engines are like magic machines that turn heat into useful energy. Our TD-8572 works using the gas law, and we can watch what happens to volume as it does its thing. This hands-on experiment helps us connect what we learn in books with real-life action, making thermodynamics easier to understand. Throughout the report, we'll talk about the basic ideas behind heat engines and gas laws, setting the stage for understanding how well our PASCO TD-8572 is doing. By looking at volume and pressure changes during the experiment, we hope to learn cool things about how efficient and effective our heat engine is. This experiment helps us dig deeper into the world of thermodynamics and how it applies to the stuff we see every day in science and engineering.

2. THE HEAT ENGINE/GAS LAW APPARATUS

The Heat Engine/Gas Law Apparatus has two ports to quickly connect to air chamber tubing. It can link to a Low-Pressure Sensor for computer interface use. The mass platform is for adding weight in experiments. The piston holding thumbscrew keeps the piston in place precisely. The millimeter scale measures how far the piston moves. Pressure port mating connectors help with pressure measurements. Two port shut-off valves control the gas flow. The rod clamp holds the engine on a stand. The precision-bore Pyrex cylinder in a plastic shield keeps gases safe for accurate observations. Together, these parts make the **PASCO TD-8572** great for hands on learning about heat engines and gas laws.

- **Base Apparatus:** The main part with a piston of radius 16.25 mm and a combined mass of 35g
- **Air Chamber:** A separate chamber for air, crucial for the functioning of the apparatus.
- **Hose Configurations:** Three types of hoses, including one with one-way check valves, one with a clamp, and a plain piece of tubing.
- **Rubber Stoppers:** One each of a single-holed and a two-holed rubber stopper.

Figure 1 and figure 2 shows each part of the apparatus of the **PASCO TD-8572 heat engine**.

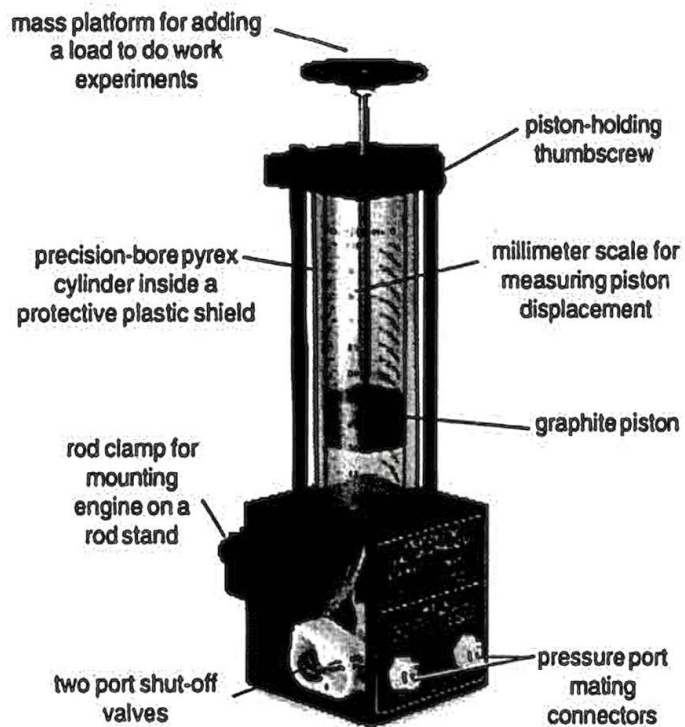


Figure 1 The PASCO TD-8572 Heat Engine/Gas Law Apparatus

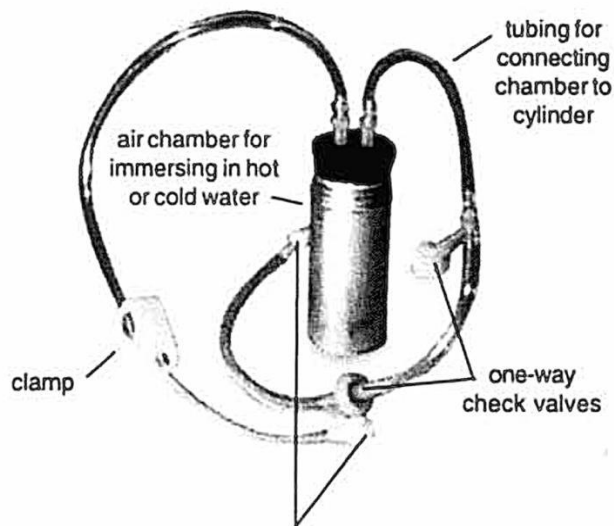


Figure 2 The PASCO TD-8572 Heat Engine/Gas Law Apparatus

3. THEORY

In our lab, we're checking out how A heat engine system behaves when temperature is altered. When temperature is altered the gas trapped inside the air chamber expands or when we can measure the work that's done.

Mechanical work: $work = Force \times distance$ or $W = F\Delta X$ in our context $W_{total} = \sum mg(y_2 - y_1)$ where y is the distance the piston has moved.

Thermodynamic work: $work = pressure \times volume$ or $W = P\Delta V$

For a thermodynamic process, we find the area under the p-v diagram

P-V Diagram: The work done (W) is the area under the curve on a Pressure-Volume (P-V) diagram:

Area under p-v graph is taken as the area of a parabola given by $(v_D - v_A) \times (P_B - P_A)$

For a piston in a cylinder,

Pressure on Piston: $P = \frac{F}{A}$ where p = pressure and A = area.

When adding a mass (m) to a vertical piston under gravity (g):

Force on Piston: $F = mg$ where m is the total mass including the piston mass and g = gravitational acceleration (9.81 m/s^2)

Therefore, Change in Pressure: $\Delta P = mg/A$

These formulas help relate force, pressure, and the area of the piston in the context of a thermodynamic process with a constant volume.

Lastly, in order to calculate the pressure values to be plotted on a P-V diagram we must add the atmospheric pressure: $P = P_{atm} + \Delta P$ where $\Delta P = mg/A$

4. EQUIPMENTS REQUIRED AND PROCEDURE:

Equipment's required:

1. Heat engine/gas law apparatus
2. 200g mass
3. 2 beakers, one to accommodate hot water and the other cold water

Procedures:

1. Trap air into the cylinder using the rubber stopper.
2. Put the air cylinder into a cold-water beaker.
3. Record the position of the piston (readings from the bottom of the piston)
4. Repeat the process until a cycle is complete:
Cycle: below are steps of processes to complete a cycle
A- Place the cylinder in cold water
B- add mass to the platform when cylinder is still in cold water
C- move the cylinder to the hot water

D- remove the mass when the cylinder is still in hot water

5. RESULTS AND DISCUSSION

Since the measured height was in millimeter, I had to convert to meters by dividing by 1000:

$$\text{e.g state A} = 4\text{mm} \times \frac{1\text{m}}{1000\text{mm}} = 0.0004\text{m}$$

for volume:

e.g state A $\text{volume} = \text{piston area} \times \text{height}$

$\text{piston area} = \pi r^2 = \pi(0.01625)^2 = 0.0008295768101\text{m}^2$ therefore,

$$\text{volume} = 0.0008295768101\text{m}^2 \times 0.0004\text{m} = 0.33183 \times 10^{-6}\text{m}^3$$

for pressure:

piston mass = 35g $\times 1\text{kg}/1000\text{g} = 0.035\text{kg}$

$$\text{e.g state A} = \text{pressure} = \frac{\text{force}}{\text{area}}$$

where force = mass $\times$ gravitational acceleration

$$= 0.035\text{kg} \times 9.81\text{m/s}^2 = 0.34335\text{ N where } 1\text{N} = 1\text{kgm/s}^2$$

$$\text{Pressure} = 0.34335\text{N}/0.0008295768101\text{m}^2 = 413.8857256\text{pa where } 1\text{pa} = 1\text{N/m}^2$$

If we Put into consideration the atmospheric pressure acting on the piston taking the atmospheric pressure to approximately equal to 101,000pa, the pressure value becomes;
 $413.8857256\text{pa} + 101,000\text{pa} = 101,413.8857256\text{pa} = 101.413886\text{ Kpa}$

The table below shows different states, height in each state, volume and pressure. The height is obtained by direct reading from the apparatus as the experiment was conducted.

Table 1 table of states, piston height, volume and pressure

state	Piston height/m	Volume/ m^3 (10^{-6})	Pressure/kpa
A	0.0004	0.33183	101.413886
B	0.0000	0	103.778947
C	0.0038	3.152391878	103.778947
D	0.0042	3.484222602	101.413886

The graph below is p-v graph done on excel that shows a complete cycle of a heat engine experiment. The graph shows the process direction as well as the two thermal reservoirs that must be present for a cycle to complete. It also shows adiabatic and isobaric processes. Q_H is the high temperature reservoir and Q_L is the low temperature reservoir. The also shows isobaric (constant pressure process) and adiabatic process (process with no heat exchange).

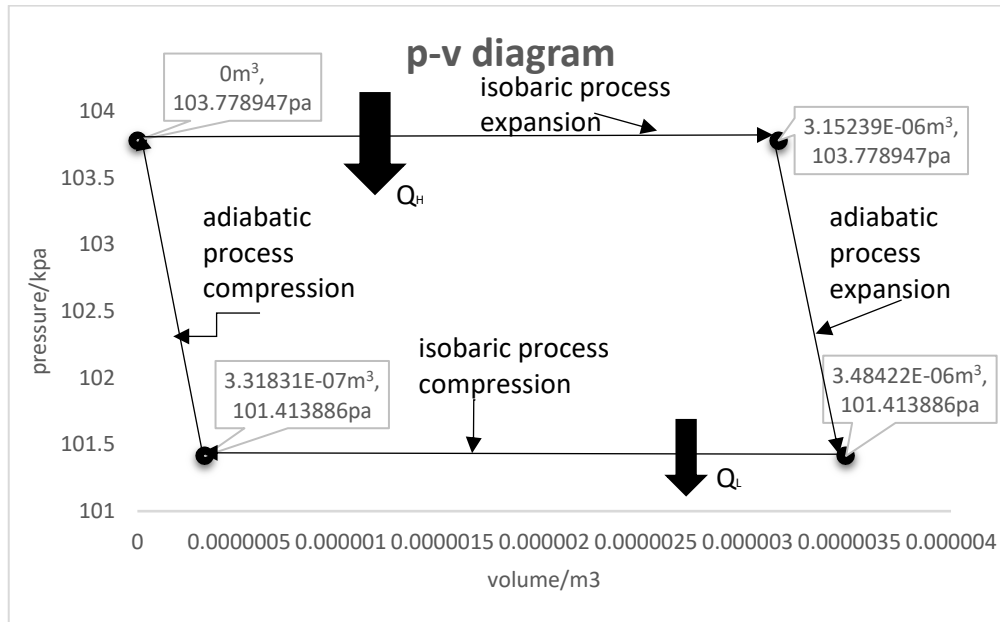


Figure 3 p-v diagram

DISCUSSION OF RESULTS

To calculate thermodynamic work, we use the formula below to get the area under the p-v graph

$$\text{Area under p-v graph} = (v_D - v_A) \times (P_B - P_A)$$

$$\text{Area} = (3.484222602 \times 10^{-6} - 0.33183 \times 10^{-6}) \text{ m}^3 \times (103.778947 \times 10^3 - 101.413886 \times 10^3) \text{ pa} = 7.4556008883 \times 10^{-3} \text{ m}^3 \text{ pa (joules), where } 1 \text{ pa} = 1 \text{ N/m}^2 \text{ so } \text{m}^3 \times \text{N/m}^2 = \text{Nm} = \text{joules}$$

Therefore, **thermodynamic work = $7.4556008883 \times 10^{-3}$ joules**

To calculate mechanical work,

$$\text{We use the formula } W_{\text{TOTAL}} = W_{\text{AB}} + W_{\text{BC}} + W_{\text{CD}} + W_{\text{DA}}$$

We recall that, adiabatic process is a process of no heat exchange and so our equation reduces to

$$W_{\text{TOTAL}} = W_{\text{BC}} + W_{\text{DA}}$$

$$W_{\text{BC}} = F \times (y_c - y_b) = 2.30535 \text{ N} \times (0.0038 \text{ m} - 0 \text{ m}) = 8.76033 \times 10^{-3} \text{ Nm} \quad \mathbf{1 \text{ Nm} = 1 \text{ joule}}$$

$$W_{\text{DA}} = F \times (y_a - y_d) = 0.34335 \text{ N} \times (0.0004 \text{ m} - 0.0042 \text{ m}) = -1.30473 \times 10^{-3} \text{ joules}$$

$$W_{\text{TOTAL}} = 8.76033 \times 10^{-3} \text{ Nm} + (-1.30473 \times 10^{-3} \text{ Nm}) = 7.4556 \times 10^{-3} \text{ joules}$$

Mechanical work = 7.4556×10^{-3} joules

We observe that both mechanical and thermodynamic work have consistent results which indicate that the methods used to calculate work are compatible. We also observe that both results are positive values indicating work is being done by the system.

our experimental investigation focused on the calculation of thermodynamic work and mechanical work in the context of a heat engine cycle. The area under the pressure-volume (P-V) graph was determined to quantify thermodynamic work using the formula Area under p-v graph = $(v_D - v_A) \times (P_B - P_A)$, resulting in a value of 7.4556008×10^{-3} joules. The calculation considered the conversion from cubic meters times pascals to joules, ensuring compatibility with the units of work.

Simultaneously, mechanical work was calculated using the formula $W_{\text{TOTAL}} = W_{AB} + W_{BC} + W_{CD} + W_{DA}$ for adiabatic processes, where W_{AB} and W_{CD} were determined to be 0 joules since there is no heat exchange in the process and so the equation for total mechanical work done reduce to $W_{\text{TOTAL}} = W_{BC} + W_{DA}$. The total mechanical work, 7.4556×10^{-3} joules, was found to be consistent with the result obtained for thermodynamic work.

6. CONCUSSION

In our experiment, we wanted to figure out how heat engines work by doing some number crunching to understand how energy changes. We used a graph that showed pressure and volume to calculate both mechanical and thermodynamic work. The results we got were consistent, which means our experiment was reliable. This helped us connect theoretical ideas like those we learn in books to real world applications. Our experiment taught us about thermodynamics principles and how heat engines operate, giving us useful information for future studies and ways to better understand how energy transforms.